

SEATauBench: Adapting Tool-Agent-User Evaluation Into Low-Resource Southeast Asian Languages

My Chiffon Nguyen¹, Aulia Adila¹, Saksorn Ruangtanusak^{1,2}, Kittiphat Leesombatwathana^{1,3},
Vissuta Gunawan Lim¹, Patomporn Payoungkhamdee^{1,4}, Samuel Cahyawijaya^{1,5}

¹SEACrowd, ²SCB DataX, SCBX Group, ³Chulalongkorn University, ⁴VISTEC, ⁵Cohere
{chiffonng136, auliaadila036, vglim3653}@gmail.com
saksorn.ruangtanusak@data-x.ai, 6534404823@student.chula.ac.th
patomporn.p_s21@vistec.ac.th, samuelcahyawijaya@cohere.com

Abstract

While AI development and evaluation for Southeast Asia (SEA) has grown rapidly, agent capabilities in regional languages are still poorly understood despite its importance to sovereign AI. To fill this gap, we introduce SEATauBench¹, the first agent-focused evaluation framework for SEA sovereign AI. SEATauBench adapts τ^2 -Bench to five languages—Mandarin, Vietnamese, Thai, Indonesian, and Filipino—and evaluates agents across progressively localized settings that vary the language of user-agent interaction, tool specifications, and task domains. Across three recent models, we find that English agent capabilities transfer reasonably well when only the conversation language changes, but quality and robustness degrade sharply as more task contexts are localized, with the largest losses in full domain adaptation. We also highlight the limits of English-only agent assessment for predicting agent capabilities in SEA languages. More broadly, SEATauBench provides a diagnostic benchmark and reusable adaptation pipeline for building reliable multilingual agents for linguistically diverse regions.²

1 Introduction

Sovereign artificial intelligence (AI) has become critical for nations seeking to maintain autonomy in their digital futures (Chae et al., 2025), including Southeast Asia. As articulated in Mushkani et al. (2025); Barasa et al. (2026), sovereign AI encompasses not only technological self-reliance but also cultural and linguistic relevance, a vital dimension for the more than 700 million people whose linguistic diversity is poorly represented by English-centric development and evaluation (Bhandari and Modi, 2026; Putra, 2024).

¹SEATauBench is pronounced "si-tau-bench", similar to the Filipino word for string beans, "sitaw".

²Data and code can be accessed at <https://github.com/SEACrowd/SEATauBench>

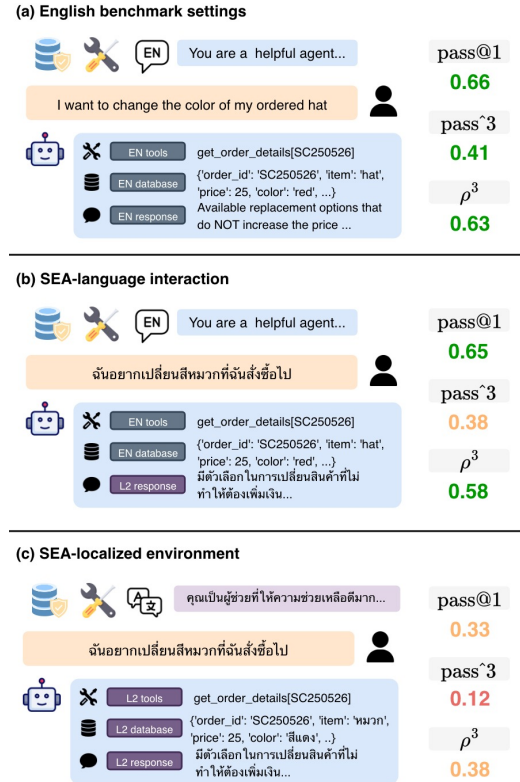


Figure 1: SEATauBench exposes a critical English-SEA agentic capability gap in existing proprietary and open-source LLMs across progressively localized evaluation scenarios. This evidence exposes the unreliability of existing English-centric benchmark to reflect the actual capabilities of LLMs for sovereign AI adoption.

SEA-specific language evaluation has expanded rapidly, with SEA-Exam and SEA-Bench (Liu et al., 2025; Zhang et al., 2024; Nguyen et al., 2024), SEA-VL (Cahyawijaya et al., 2025a, 2026), SEACrowd (Lovenia et al., 2024), SEA-HELM (Susanto et al., 2025), NusaCrowd (Cahyawijaya et al., 2023a), NusaWrites (Cahyawijaya et al., 2023b), and NusaX (Winata et al., 2023) establishing valuable foundations for measuring regional language understanding, cultural knowledge, reasoning, safety, and multimodal capabilities. Nevertheless, these benchmarks primarily

assess static model behaviors, with limited work evaluating whether agents can complete multi-turn, tool-mediated tasks in regional languages, the capabilities required when more and more AI systems operate in real-world deployments like service, commerce and travel (Budzianowski et al., 2018; Eric et al., 2020; Zang et al., 2020; Han et al., 2021; Ye et al., 2022).

To address this gap, we introduce SEATauBench, the first agent-focused evaluation framework for SEA. SEATauBench adapts τ^2 -Bench (Yao et al., 2024; Barres et al., 2025; Shi et al., 2026; Ray et al., 2026) into five target languages L2 (Mandarin Chinese, Vietnamese, Thai, Indonesian, Filipino) and evaluates agents across three progressively localized settings: (1) **L2 Interaction**, isolating linguistic capability in user-agent conversation; (2) **L2 Tool**, testing the ability to use tools with non-English tool specifications; and (3) **L2 Domain**, evaluating performance when all task contexts are in L2 (Section 3.3). To translate different interfaces an AI agent interacts with, without breaking the execution of τ^2 -Bench, we develop a structured, non-breaking translation pipeline (Section 3.2).

Across three recent agent models, we find that English agent capabilities transfer reasonably well when agents only need to respond in target languages, but quality and robustness degrade sharply once tools, policies, and task contexts are progressively provided in SEA languages (Section 5.1). These results expose a gap between the growth of SEA evaluation resources and the readiness of current agents for sovereign AI deployment, establishing SEATauBench as a diagnostic benchmark for building reliable multilingual agents for the region.

2 Related Works

2.1 Agent Evaluation

Task-oriented dialogue benchmarks such as MultiWOZ (Budzianowski et al., 2018; Eric et al., 2020; Zang et al., 2020; Han et al., 2021; Ye et al., 2022) and MASSIVE (FitzGerald et al., 2023) established evaluation for goal-directed dialogue and multilingual intent-slot understanding. Recent tool-use benchmarks, including ToolEval (Qin et al., 2024) and BFCL (Patil et al., 2025), extend this direction to function calling and API use. However, deployed service agents must further sustain multi-turn interactions, follow domain policies, and update external states. τ^2 -Bench (Yao et al., 2024; Barres et al., 2025; Shi et al., 2026; Ray et al., 2026)

address this through simulated user-agent-tool environments, scored by task completion metrics pass¹ and pass³. Related benchmarks further assess realistic agent behavior in business and professional settings (Huang et al., 2025; Drouin et al., 2024; Boisvert et al., 2024; Xu et al., 2026; Patwardhan et al., 2026).

Despite these advances, existing agent benchmarks remain largely centered on high-resource languages, limiting their applicability to sovereign AI settings. SEATauBench addresses this gap by extending τ^2 -Bench to SEA languages through localized conversational and tool-use scenarios. To our knowledge, it is the first multilingual benchmark to preserve the task-oriented evaluation framework of τ^2 -Bench, contributing towards reliable assessment of agent capabilities in multilingual contexts.

2.2 Multilingual Evaluation

Multilingual evaluation has mainly targeted understanding, reasoning, translation, and multimodal comprehension through benchmarks such as mMMLU (Hendrycks et al., 2021; OpenAI, 2024), GlobalMMLU (Singh et al., 2025), GlotEval (Luo et al., 2025), CVQA (Romero et al., 2024), and AyaVisionBench (CohereLabs, 2025). For South-east Asian languages, resources such as NusaX (Winata et al., 2023), NusaWrites (Cahyawijaya et al., 2023b), NusaDialogue (Purwarianti et al., 2025), IndoTOD (Kautsar et al., 2023), SEACrowd (Lovenia et al., 2024), StingrayBench (Cahyawijaya et al., 2025b), SEAExam/SEABench (Liu et al., 2025), SEA-HELM (Susanto et al., 2025), SEA-VL (Cahyawijaya et al., 2025a), and SEA-VQA (Urailertprasert et al., 2024) expand evaluation to local linguistic, cultural, reasoning, safety, and multimodal contexts. However, these benchmarks largely focus on static evaluation rather than interactive tool use or multi-turn task completion.

The closest work is MASSIVE-Agents (Kulkarni et al., 2025), which evaluates function calling across 52 languages using BFCL, but remains limited to function selection and argument prediction. SEATauBench goes further by evaluating full user-agent-tool interaction, requiring agents to interact over multiple turns, follow instructions, use tools, and complete realistic tasks in SEA languages. It therefore bridges multilingual function calling and interactive agent evaluation, addressing a gap left by existing SEA multilingual and agent evaluation.

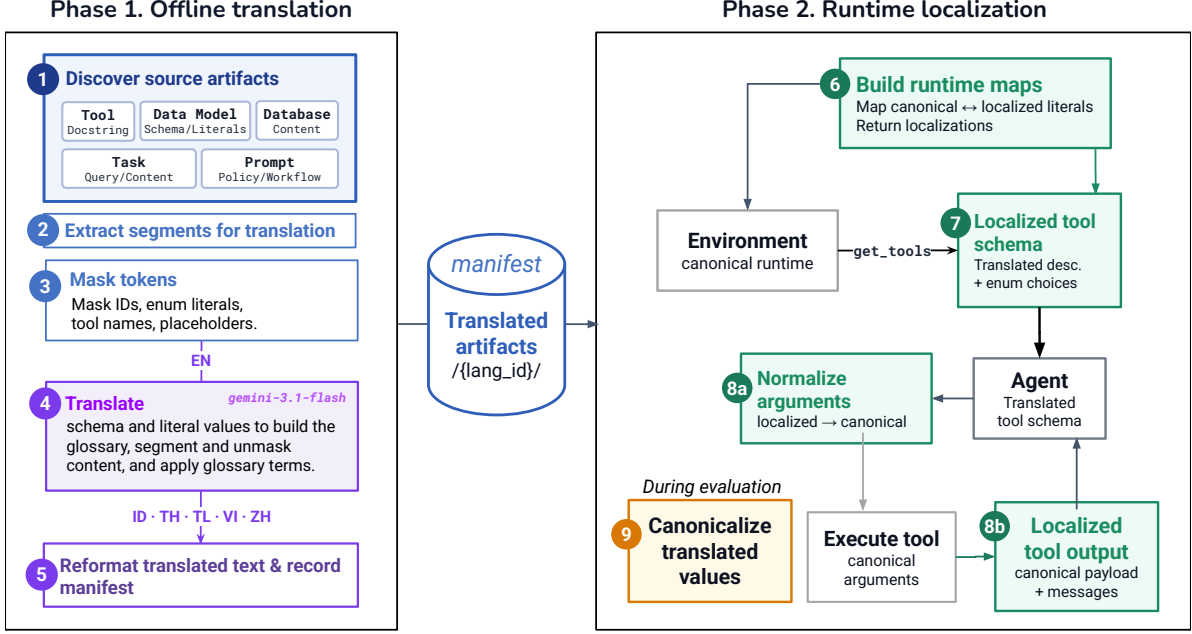


Figure 2: Overview of the automated translation pipeline for generating multilingual τ^2 -Bench artifacts (Section 3.2). We provide more details about the pipeline and the resulting translated artifacts in Appendix A.

3 Adapting τ^2 -Bench to SEATauBench

3.1 Background

We construct SEATauBench by extending the English tool-agent-user benchmark τ^2 -Bench (Barres et al., 2025) to multilingual settings. This adaptation must account for two coupled agent-facing surfaces: **interaction content** (task definitions, domain policies and workflows, and structured databases) and **executable interfaces** (tool schemas and, in telecom, return messages).

At runtime, the agent must read policies, reason about tasks, call tools, inspect outputs, and talk to a simulated user in the target language. The benchmark therefore needs a localized interface whose visible text is translated, while the execution layer still preserves canonical English values. Our pipeline aligns these representations to preserve task semantics and metric comparability.

Section 3.2 describes how we translate and construct L2 artifacts, and Section 3.3 uses those artifacts to define controlled evaluation scenarios with increasing levels of L2 adaptation.

3.2 SEATauBench L2 Adaptation Pipeline

Figure 2 summarizes our two-phase adaptation pipeline: **offline translation**, which materializes language-specific assets, and **runtime localization**, which patches the environment observed by the

agent. The full details are provided in Appendix A.

We adapt three domains from τ^2 -Bench (retail, airline, and telecom) into five L2: Vietnamese (vi), Indonesian (id), Thai (th), Filipino (tl), and Mandarin Chinese (zh).³

Offline Translation. We translate static domain assets to construct L2 artifacts. It extracts natural-language spans from tasks, policies, databases, tool docstrings, schemas, and tool-return templates, while masking executable tokens such as IDs, status values, tool names, and structural markers. It first translates schema literals to establish a glossary, preventing the same executable value from being rendered inconsistently. The pipeline then writes outputs with format-specific writers and records per-language manifests with model metadata and source-file SHA-256 fingerprints. Appendix A.5 reports the resulting artifact statistics.

Runtime Localization. The runtime localization phase handles dynamic content exposed during inference. First, it localizes the tool schema shown to the agent: descriptions, enum choices, and examples are rendered in L2 so the scenario tests target-language tool use rather than English schema reading, while the underlying implementation remains unchanged. Second, it preserves executability by

³We do not include other SEA languages such as Malay, Lao, or Cambodian due to limited annotator capacity.

Scenario	L2 Tools	L2 Convo	L2 DB & Policy
(S1) English Only	✗	✗	✗
(S2) L2 Interaction	✗	✓	✗
(S3) L2 Tool	✓	✗	✗
(S4) L2 Domain	✓	✓	✓

Table 1: SEATauBench evaluation scenarios across English and five SEA languages. S2 and S3 isolate L2 dialogue and L2 tool docstring respectively, while S4 L2 domain uses all translated contexts.

normalizing localized arguments back to canonical English values before tool calls; otherwise, correct L2 arguments could fail only because the original tools expect English literals. After execution, tool responses are localized back into L2, and final payloads are canonicalized again before scoring, keeping the interaction monolingual for the agent while making metrics comparable across languages and with the English benchmark. Together, offline translation and runtime localization let us vary which benchmark surfaces are exposed in L2, forming the scenario design.

3.3 Evaluation Scenarios

SEATauBench has four scenarios with increasing level of L2 adaptation (summary in Table 1).

(S1) English Only. The original English benchmark, to establish the baseline performance for tested agents.

(S2) L2 Interaction. The simulated user and agent converse in L2, while tools, policies, databases, and task contexts remain in English.

(S3) L2 Tool. Tool schemas are rewritten in L2 while conversation and domain context remain in English. We present translated tool schemas in two formats: single-L2, where each schema uses one target language, and mixed-L2, where schemas combine an increasing number of target languages.

(S4) L2 Domain. Agents operate in a complete L2 setting where dialogue, tool schemas, policies, task descriptions, and agent-visible database are translated while task semantics remain fixed. This scenario uses all translated artifacts and runtime localization described in Section 3.2.

4 Experimental Setup

Metrics. Following τ^2 -Bench (Barres et al., 2025), we report two task metrics. First, $\text{pass}@1$ measures quality as the mean success rate of inde-

pendent trials reaching the expected final state, i.e., $\text{pass}@1 = \text{pass}^1 = \mathbb{E}[r]$. Building on existing metrics, we also derive a robustness measure, $\rho \in [0, 1]$, defined as

$$\rho^q = \frac{\text{pass}^q}{\text{pass}@1} \quad (1)$$

where ρ^q denotes the probability that all q independent trials succeed, averaged across tasks. $\rho^q = 1$ indicates that the agent performs consistently across multiple runs, while lower ρ^q indicates that the agent may solve a task once but fail to do so reliably across repeated trials.

Task and Domain. For high-quality evaluation, we use latest versions of retail and airline domains (Cuadron et al., 2026), which incorporate several corrections and refinements over prior versions (Yao et al., 2024; Barres et al., 2025). For the telecom domain, we follow the original τ^2 -Bench, as its tasks were not updated in the most recent release. We leave exploration of the banking domain (Shi et al., 2026) and the voice modality (Ray et al., 2026) introduced in τ^3 -bench to future work.

Hyperparameters and Models. For all evaluations, we use $q = 3$, resulting in the robustness metric ρ^3 , and Qwen3-235B-A22B-Inst⁴ (Qwen-Team, 2025) as the user LLM. For the natural-language assertion judge, we use GPT-4.1, following the implementations provided in τ^2 -Bench. We evaluate three recent LLM agents spanning three model families: GPT-5-Mini⁵ (Singh et al., 2026) as a proprietary model, Qwen3-235B-A22B-Instruct-2507 as a representative open-source model, and Kimi-K2.5 (Kimi Team et al., 2026) as a larger-scale open-source model.

For each model, we use the default hyperparameters defined in each provider (See Appendix C). We avoid any model-specific hyperparameter tuning to better reflect the out-of-the-box tool-agent-user interaction capability when adopted to the specified scenario and L2.

5 Results

We present two main results. First, we examine how quality and robustness change as more components of the English benchmark are converted to L2 (Section 5.1). Second, we test whether these trends are consistent across agent models and task domains (Section 5.2). Together, the results show

⁴<https://huggingface.co/Qwen/Qwen3-235B-A22B-Inst-A22B-Instruct-2507>

⁵<https://developers.openai.com/api/docs/models/gpt-5-mini>

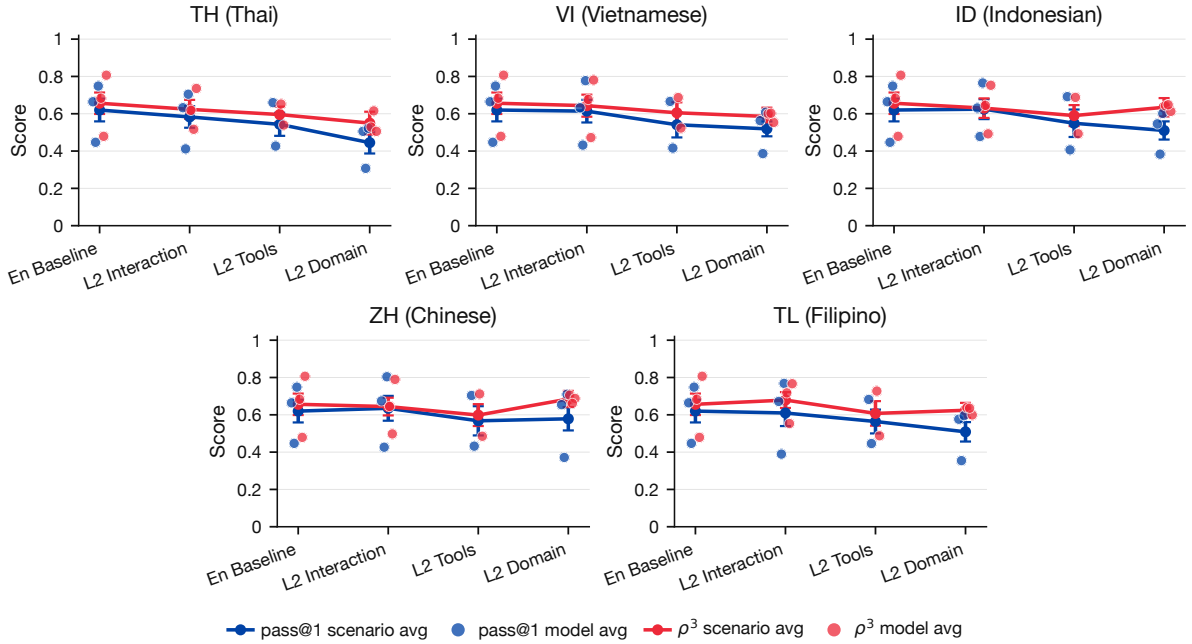


Figure 3: Evaluation results on SEATauBench. SEATauBench reveals consistent degradation as the setting becomes less English-centric on all languages in both pass@1 and ρ^3 metrics, with most severe drops in low-resource languages with non-Latin scripts. All scenario results are reported with three models, except for L2 Tool, where only GPT-5-mini and Qwen3-235B-A22B-Inst are reported.

that English-centric evaluations overestimate multilingual agent reliability, and that the size of this gap depends on both the target language and the structure of the task environment.

5.1 Performance Degrades in L2 Languages

Figure 3 demonstrates a consistent decline in model performance as the evaluation setting becomes progressively less English-centric. Both pass@1 and ρ^3 are strongest in English settings, but performance drops when the user-agent conversation is conducted in an L2, drops further when tool interfaces are localized, and typically reaches its lowest levels when the domain context is also translated. This progression indicates that multilingual agentic competence is not only a matter of understanding non-English user utterances. Agents must also use localized tool schemas, entities, and ground decisions in domain policies expressed in L2.

Although all L2 settings degrade relative to English, the severity and form of degradation differ by language. In L2 Domain, Thai and Filipino show the largest pass@1 declines, falling to roughly the low-0.4 range. Vietnamese and Chinese remain closer to the mid-0.5 range, while Indonesian shows a partial robustness recovery despite lower pass@1. The sharper drop in Thai may re-

flect challenges associated with a distinct writing system and lower-resource coverage. However, the comparable drop in Filipino shows that localization failures are not limited to non-Latin scripts: even Latin-script languages can expose weaknesses when agents must reason over fully localized dialogue, tools, schemas, and domain context.

Quality and robustness diverge. Figure 3 also highlights that pass@1 and ρ^3 do not always move together. In Indonesian, Filipino, Chinese, and to a lesser extent Vietnamese, pass@1 drops in L2 Domain while ρ^3 remains comparatively stable or rebounds relative to L2 Tool. This means that agents may solve fewer tasks overall, but when they do solve a task, their successful behavior is relatively reproducible in some L2 settings. Thai is the clearest exception: both pass@1 and ρ^3 decline in L2 Domain. This suggests that some L2 conditions weaken not only average task success but also consistency across repeated trials.

5.2 Does the Trend Hold across Agents?

Figure 4 shows that model performance differs across metrics. For pass@1, Kimi K2.5 performs best in most languages, including English, Vietnamese, Filipino, and Chinese, while GPT-5 Mini leads in Indonesian and Thai, indicating that model

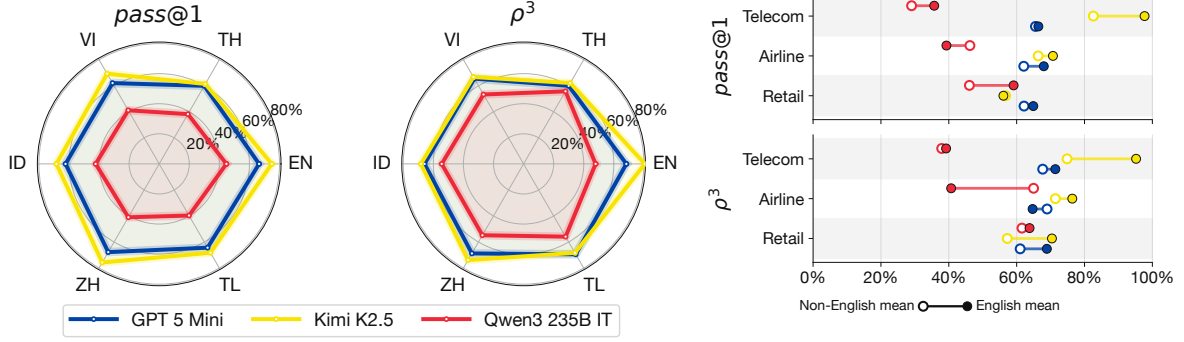


Figure 4: Quality and robustness of varying agent models across languages and domains. **(left)** All agents yield consistently higher $pass@1$ in English and Chinese indicating a quality bias toward high-resource languages. **(center)** A similar trend is observed on ρ^3 , with negative correlation of ρ^3 between English and L2 languages especially Thai. **(right)** The trend of $pass@1$ holds across most domains and models with some exceptions as described in §5.2.

advantages do not transfer uniformly across SEA languages. Qwen3-235B-A22B-Inst consistently has the lowest $pass@1$, suggesting weaker single-run task completion. However, ρ^3 reveals a different pattern: despite its lower $pass@1$, Qwen3-235B-A22B-Inst often shows stronger robustness in lower-resource L2 settings. This robustness advantage weakens in higher-resource languages such as English and Chinese, where Kimi K2.5 and GPT-5 Mini remain competitive or stronger.

The domain-level results show that the English–non-English gap varies across domains and models. For GPT-5 Mini and Kimi K2.5, $pass@1$ generally follows Telecom > Airline > Retail in both English and non-English settings, suggesting that Retail is the most challenging domain due to its policies, item-level constraints, and state updates. Qwen3-235B-A22B-Inst Instruct shows a different pattern: Retail performs best in English but becomes weakest in non-English settings, indicating that domain-specific strengths in English may not transfer under localization. The ρ^3 results further show that localization reduces robustness, with English settings generally achieving higher consistency than non-English settings across domains. However, this degradation is not uniform. For Qwen3-235B-A22B-Inst, agent performance follows the expected non-English decline in Telecom and Retail domains, while shows a relative robustness spike for Airline. This suggests that robustness under localization depends on both task domain and the agent model behavior.

6 Analysis and Discussion

In this section, we examine error patterns arising from conversational user–agent interactions in L2

Interaction and L2 Domain scenarios (Section 6.1) and the effects of having tools in multiple languages (Section 6.2). Furthermore, we show that agent performance has little to no association with language correctness (Section 6.3) and that English performance is an unreliable proxy for L2 performance with SEA languages (Section 6.4).

6.1 Error Analysis

To further examine the downstream performance metrics reported in the previous section, we conduct a qualitative error analysis based on the LLM-as-a-Judge framework proposed in Shi et al. (2026). Following Barres et al. (2025), we categorize task simulation errors into two types: *Critical* error, in which the agent deviates from the user’s intent or commits irrecoverable mistakes, and *Benign* error, which do not prevent successful task completion. For early results, we analyze only scenarios that affect both behavioral changes on both simulated user and agent, i.e., L2 Interaction and L2 Domain scenarios, since L2 Tool only translates tool specifications while keeping all tool-agent-user interactions in English. Specifically, we employ DeepSeek-V4-Flash as the oracle LLM on simulations done by GPT-5-mini as the agent and Qwen3-235B-A22B-Inst as the simulated user.

L2 Interaction. Agent performance degrades when interacting with users in L2. As shown in Figure 5, critical errors appear in around 40% of total simulations across SEA languages, with Thai being the most severe (nearly 50%). Moreover, when the user model must communicate only in SEA languages, its behavior becomes less stable and less consistent with the intended simulation



Figure 5: Error categorization for simulated user and agent, across SEA language in the L2 Interaction and L2 Domain scenarios. Both agents and users tend to have higher critical errors in L2 than English, while having a similar benign error in comparison to English. The errors are reported by judge DeepSeek-V4-Flash, for simulated user Qwen3-235B-A22B-Inst and agent GPT-5-mini.

setup, with the most severe degradation in Filipino.

L2 Domain. Agent performance degradation is more pronounced in the L2 domain adaptation setting. There are slightly more critical errors for agents here than L2 interaction, with clearest trends in Thai, Filipino, and Indonesian (Figure 5). This finding suggests that operating in complete L2 environments further impairs the agent’s ability to complete tasks effectively.

Simulated User. A similar trend is observed on the simulated user, where critical errors consistently account for around 20% of all errors in both analyzed scenarios. Our main finding that agent performance drops as L2 adaptation increases across four scenarios (Figure 3) might be compromised by the low reliability of the simulated user. This account previously appeared in Barres et al. (2025) for English and Seshadri et al. (2026) for dialectal English.

6.2 Mixed language tool causes modest drop in task performance

Beyond adaptation using a single L2, we also investigate how agents handle tools in multiple languages, which is increasingly relevant as tools for agents are developed in different world regions. We do so by extending L2 Tool experiment to five different mixed language settings of tool specifications, by progressively adding more languages to

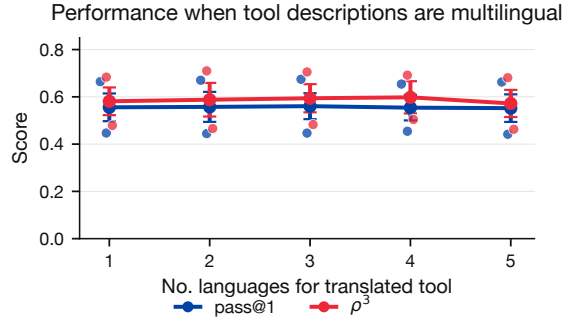


Figure 6: Pass@1 and ρ^3 as the number of languages used for translated tools increase from 1 (English) to 5. Performance for two agents (GPT-5-Mini & Qwen3-235B-A22B-Inst) slightly drops when a second language is added, but it plateaus for more languages.

the mix: Mix-2 adds Thai to English, Mix-3 adds Vietnamese, Mix-4 adds Indonesian, and Mix-5 adds Chinese (for more details, see Appendix B).

Figure 6 shows that mixed language tool causes only a modest drop. Across all mixed-language settings, average pass@1 drops from 0.68 for English-only to 0.55 (0.13 decrease). ρ^3 shows a similar trend of degradation (0.11 decrease). These results indicate that agents with strong English tool-using performance might use tools only slightly less effectively when tools are multilingual; however, increasing the language diversity in tool specification has little to no impact to the overall agentic tool-

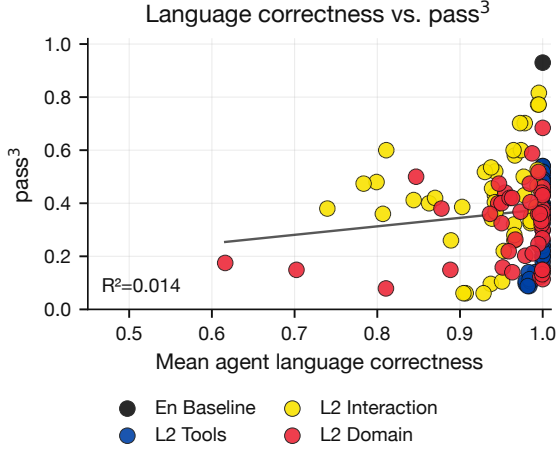


Figure 7: Only 1.4% of the variation in task performance metric pass^3 is explained by language use capability of tested agents, suggesting a very weak linear relationship. Each dot represents a run for one scenario $\times$ domain $\times$ language setting.

using capability.

6.3 Language use is uncorrelated with performance

We also measure whether the simulated user and agent operated across tasks in the correct L2, or *language correctness*. It is calculated as the fraction of eligible turns whose detected language (by fast-Text) matches the expected L2 for a single run. We exclude trajectories with system errors and other errors unrelated to task execution.

Figure 7 shows that run-level language correctness for agent has little association with task performance: a linear fit explains only $R^2 = 0.014$ of the variance in pass^3 . Several runs achieve near-perfect language correctness while spanning a wide range of pass^3 scores, especially in L2 Tool and L2 Domain. Conversely, L2 Interaction contains runs with visibly lower language correctness but still moderate or high task success. This suggests that, at the aggregate run level, task failures are not primarily explained by whether the agent remains in the target language. For language drift, we also measure how often the agent produces text in the expected L2 (Figure 9), where in a conversation off-target text appears (Figure 10), and how much of the off-target text is specifically English (Figure 11).

6.4 Is English a reliable proxy for L2 in SEA?

The τ^2 -Bench evaluation framework, while effective for measuring agent capabilities, is resource-

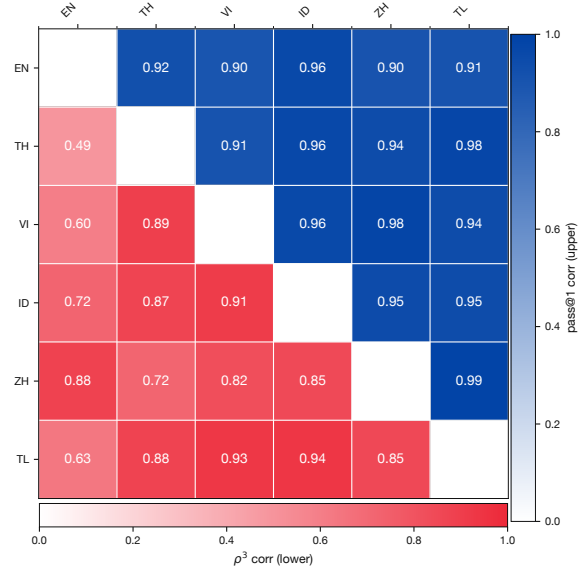


Figure 8: Correlation of $\text{pass}@1$ (upper triangle) and ρ^3 (lower triangle) across English and SEATauBench languages. Correlation scores are aggregated by four scenarios, three agent models, and three task domains.

intensive and computationally expensive. Extending to SEATauBench with an additional language dimension adds to these costs. To address this challenge, we examine correlation between quality and robustness metrics across SEATauBench languages to determine whether performance in one language could serve as a proxy for other languages.

As illustrated in Figure 8, agent performance in English has high predictive power of L2 performance for $\text{pass}@1$, with correlation above 0.9. However, there is more variability in ρ^3 , where the highest correlation is in Mandarin Chinese (0.88) and the lowest in Thai (0.49). Therefore, English performance is not a reliable representation for SEA language performance when we factor in consistency. Filipino (TL), however, shows the strongest correlation with other SEA languages, with the lowest score being 0.91 for $\text{pass}@1$ and 0.85 for ρ^3 . Based on these findings, we recommend using the Filipino subset of SEATauBench as a proxy to estimate performance across the remaining four languages, for a more efficient yet representative evaluations of AI agents in the SEA region.

7 Conclusion

SEATauBench establishes the first agent-focused framework for SEA linguistic diversity, revealing that English agentic capabilities transfer effectively

to a superficial L2 interaction but degrade significantly with the inclusion of non-English contexts, due to both agent limitations and reduced user capacity. By exposing fundamental gaps in English-only evaluation methodologies as a proxy of agents capability in SEA languages, SEATauBench provides essential diagnostic tools for sovereign AI development, establishing a rigorous foundation for future research in sovereign agentic AI solutions, ultimately supporting autonomous AI across diverse SEA language communities.

8 Limitation

This work presents a systematic framework for extending agentic benchmarks to evaluate multilingual conversational capabilities with tool access in languages spoken in SEA, specifically Vietnamese, Indonesian, Thai, Filipino, and Chinese, across the retail, airline, and telecom domains. Although the proposed framework can, in principle, be extended to additional languages, the empirical findings and their implications are confined to English and SEA languages covered in this work, and may not generalize to other languages.

Another limitation is the benchmark adaptation methodology, which is specifically designed around the τ -Bench evaluation paradigm (Yao et al., 2024; Barres et al., 2025). While the proposed evaluation scenarios and translation pipeline may be applicable more broadly, the underlying design constraints may limit the generalizability of the methodology beyond the τ -Bench frameworks. We encourage others to reproduce our findings, and extend our proposed methodology and analysis to other languages and agent evaluation.

9 Ethical Statement

We conducted this research with careful attention to ethical considerations throughout the development process. In developing of the translation pipeline and the evaluation framework, we ensure that there is no external sensitive information is exposed throughout the process and the resulting artifacts. We recognize our responsibility to ensure these systems are fair, respectful of different cultures, and beneficial to all users. Our approach focuses on being transparent about what the agents can and cannot do, actively working to identify and address potential biases, and taking steps to prevent misuse. We handle all research data responsibly, protecting privacy while respecting linguistic di-

versity. This work aims to contribute to AI that serves people equitably across languages and cultures, and we are committed to being accountable for how these technologies impact society.

Acknowledgment

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A Machine Translation Pipeline Details

A.1 τ^2 -BenchContext

Each τ^2 -Benchdomain (airline, retail, telecom) exposes the agent to two coupled surfaces. First, `data/tau2/domains/{domain}/` contains the benchmark content seen during an interaction: task definitions in `tasks.json`, domain policies and workflow documents in markdown files `*.md`, and structured databases in `db.json`, `db.toml`, or user-side DB files. Second, `src/tau2/domains/{domain}/` contains the executable tool interface: toolkits in `tools.py` and `user_tools.py`, schema definitions in

data_model.py and user_data_model.py, and in telecom, tool return string templates.

At runtime, the agent must read policy, reason about tasks, call tools, inspect tool outputs, and communicate with a simulated user entirely in the target language.

A.2 Translation Principles

Adapting τ^2 -Bench to non-English languages requires translating the **natural-language content** seen by the language model while leaving untouched the **runtime-canonical tokens** consumed by the tool execution layer: enum literals, entity identifiers, status codes, and tool argument names. Conflating these two layers would corrupt evaluation: a translated product name that reaches the execution engine might trigger a lookup failure, while an untranslated persona description breaks the monolingual interaction we intend to test. We therefore build the pipeline around four principles: **Runtime invariance**. Canonical tokens remain English at execution time. Only the LLM-visible surface is localized.

Terminological consistency. A canonical value should surface with the same target-language realization everywhere it appears.

Format fidelity. Translation is applied only to natural-language leaves. Structural keys, numeric fields, booleans, and program identifiers are never modified.

Bidirectional transparency. Localized values can be mapped back to canonical English forms before execution and before metric computation.

The implementation has two phases: an **offline translation phase** that materializes language-specific assets, and a **runtime localization phase** that patches the environment presented to the agent.

A.3 Phase 1. Offline Translation

The offline phase consumes a domain’s static artifacts, translate with **Gemini 3.1 Flash Lite**, and writes translated outputs to data/tau2/domains/{domain}/{lang_id}/.

Step 1. Artifact discovery For each selected domain, the pipeline discovers files from data/tau2/domains/{domain}/ and src/tau2/domains/{domain}/, then assigns each file a processing kind (markdown, json, toml, or python). Refer to Table 2 for the list of artifacts and their role in translation.

Step 2. Segment extraction with path-sensitive rules. Each file is reduced to minimal translatable segments together with metadata describing its source path and type.

- Markdown files yield a single full-document segment.
- Task JSON is walked recursively, but only a curated allowlist of paths is translated, including persona fields, task instructions, reason-for-call, natural-language assertions, and user-or assistant-visible message history.
- Database JSON/TOML is translated only at conservative natural-language leaf keys such as name, title, description, summary, and notes. Domain-specific additions may extend this set: the airline domain also translates address1, address2, and city (user profile address text that is natural language and safe to localize).
- Tool Python files are parsed with ast; only docstrings attached to @is_tool or @is_discoverable_tool methods are extracted.
- Google-style tool docstrings are decomposed into short description, long description, parameter descriptions, returns text, and raises text so that each part can be translated independently and later reassembled.
- Schema Python files are converted into JSON artifacts whose translatable fields include class descriptions, field descriptions, enum values, and Literal[...] alternatives.
- Tool return files expose both exact response strings and template strings as translatable segments.

Step 3. Canonical-token masking. Before translation, the pipeline collects and masks strings that must remain canonical: IDs such as order_* or booking_*, status values, structural task markers, tool names, and docstring section headers such as Args or Returns. Protection is partly global and partly contextual. For example, airline literals such as basic_economy or round_trip are protected when they occur in cabin-class or trip-type contexts, but not when similar surface strings appear as ordinary prose. The masking layer replaces protected strings with opaque placeholders and restores them after translation.

Artifact Class	Source Files	Translatable Content	Runtime Alignment Purpose
Task setup	tasks.json	Persona text, instructions, reason-for-call, history	Defines user personas and evaluation-facing assertions.
Policy & workflow	*.md	Full document prose	Governs the agent’s allowed behavior and procedures.
Tool descriptions	tools.py, user_tools.py	Docstrings of tool-decorated methods	Conditions how the agent interprets tool semantics.
Schemas & literals	data_model.py, user_data_model.py	Descriptions, enum values, and Literal[...]	Defines choices appearing in dynamic tool schemas.
Database content	db.json, db.toml	Natural-language leaves (e.g., names, notes)	Supplies textual fields surfaced through tool results.
Tool return messages	tool_returns.json	Exact messages and parameterized templates	Ensures dynamic strings returned by tools remain localized.

Table 2: Artifact classes processed during offline translation and their role in maintaining runtime evaluation alignment.

Step 4. Schema-first literal translation.

Schema enum labels and `Literal[...]` values are translated first in a dedicated literal mode. This produces language-specific schema artifacts from which the pipeline builds a domain literal map, including alias forms such as underscore, space-separated, or hyphenated variants of the same canonical value. This step fixes the localized terminology before any longer-form prose is translated.

Step 5. Standard translation with glossary injection.

All remaining segments, including task prose, policy text, tool docstrings, database leaves, and tool-return templates, are translated in standard mode. The model receives the schema-derived literal map as a glossary, and segments are pre-masked so that localized forms are restored consistently. Requests are deduplicated when possible, batched as structured JSON, and executed concurrently through LiteLLM. The current implementation requires the exact Vertex route `vertex_ai/gemini-3.1-flash-lite-preview`. Batch failures can be retried, recursively split, or rerun individually if placeholder restoration fails.

Step 6. Format-aware writing and manifest recording.

The translated text is written back using format-specific writers: markdown is emitted directly; JSON/TOML files are patched only at extracted addresses; tool docstrings are reconstructed and saved as `tools.json` or `user_tools.json`; schema artifacts are written as `data_model.json` and `user_data_model.json`; and DB files with no translated leaves are still copied through so the translated directory remains complete. Each language directory also receives `translation_manifest.json`, which records the

output file, component, source language, target language, model, translation timestamp, and SHA-256 fingerprints of the source files. This manifest enables later staleness checks without forcing retranslation.

A.4 Phase 2. Runtime Localization

The offline artifacts cover static content, but the agent also sees dynamic tool schemas and tool outputs at inference time. These are handled by a runtime localization layer built from the paired source and translated schema artifacts.

Step 7. Build runtime localization maps.

A `SchemaRuntimeLocalizer` constructs four resources from the source and localized schema artifacts: a description map, a canonical-to-localized literal map, a localized-to-canonical inverse map, and optional maps for exact and templated tool-return localizations.

Step 8. Localize the tool schema shown to the agent.

The environment’s `get_tools()` path is wrapped so that the agent receives localized tool schemas: descriptions are translated, enum choices are shown in the target language, and default or example literal values are localized where appropriate. The tool implementation itself is unchanged.

Step 9. Normalize localized arguments and localize tool outputs.

Before tool execution, localized enum arguments supplied by the agent are mapped back to their canonical English values. After execution, structured response payloads and tool-return messages are localized back into the target language so the interaction remains monolingual from the agent’s perspective.

Step 10. Canonicalize localized values for evaluation. Prior to metric computation, localized payloads are canonicalized back to English, making pass-rate comparisons directly comparable across languages and against the original English benchmark.

A.5 Translated artifact statistics

Tables 3, 4, 5, and 6 provide summary statistics for translated artifacts resulting from offline translation phase (Appendix A.3).

Domain	Files / Language	Total Translated
Airline	5	25
Retail	5	25
Telecom	13	65
Total	23	115

Table 3: No. static artifact files translated per domain. Each domain is translated into 5 target languages.

Domain	Tasks	Docstrings	Policies	Return Msgs
Airline	250	14	5	0
Retail	570	16	5	0
Telecom	570	43	25	60

Table 4: No. instances per type translated across all 5 target languages.

Domain	Total Models	Value Sets	Localized Values
Airline	23	15	21
Retail	15	6	14
Telecom	18	12	51

Table 5: Schema artifacts and literal inventories.

B Detail on Mixed Language Tool Adaptation

For the mixed language tool adaptation, our goal is to investigate the stability of agents when handling tools in mixed languages. We conducted a similar experiment to the (S3) L2 Tool Adaptation where we modify the language of the tool specification into L2, but instead of converting to a specific L2, in this experiment we convert the tool specifications to a mix of several L2 languages. Specifically, we explore 5 different mixed language settings by gradually increasing the number of languages in the tool specification while maintaining a similar composition across domains: Mix-2 uses English and Thai, Mix-3 adds Vietnamese, Mix-4 adds Indonesian, and Mix-5 adds Chinese. For each run,

Domain	Collections	Record Breakdown
Airline	3	flights: 300, users: 500, reservations: 2k
Retail	3	products: 50, users: 500, orders: 1k
Telecom	6	plans: 5, devices: 29, lines: 9, customers: 4, bills: 6

Table 6: Database artifacts. Structure is preserved; only designated leaf fields are translated.

the language of a tool is fixed for the entire run to reduce noise from random sampling. For instance, if `get_item` tool is assigned to Thai, it remains in Thai across all examples. For each non-English language added, three tools are localized into that language, while the remaining tools stay in English; thus, the English-tool count decreases as more languages are added.

C Hyperparameters

Hyperparameter	Agent LLM	User Sim. LLM
Temperature	0.0	0.0
Nucleus sampling (p)	1.0	1.0
Max. generation tokens	—	—

Table 7: Inference hyperparameters for the L2 Interaction Setting. Values follow the τ^2 -Bench defaults; parameters not listed are left at the provider model defaults.

All models in the L2 Interaction Setting are run with the default inference parameters from τ^2 -Bench, as shown in Table 7.

D Compute Budget

Scenario	Agent LLM Cost (USD)
English Baseline	\$70.40
L2 Tool Adaptation	\$340.73
L2 Interaction	\$403.27
L2 Domain Adaptation	\$440.43
Total	\$1,254.83

Table 8: Approximate agent LLM inference cost (USD) for each experimental scenario, aggregated across three evaluation trials and estimated from token usage with published API pricing.

For agent model inference, we use two models across all experimental settings: GPT-5-mini accessed via the OpenRouter API, and Kimi-K2.5

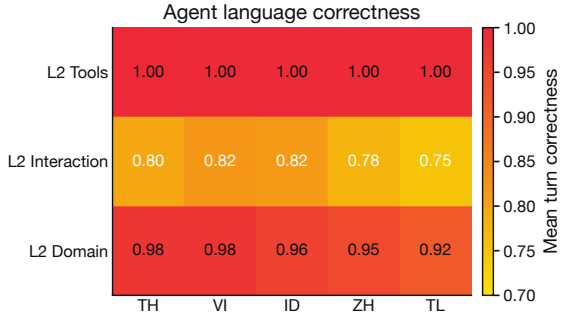


Figure 9: Average turn-level agent language correctness by setting and L2. Language correctness is calculated as the fraction of eligible turns whose detected language (by fastText) matches the expected L2 for one turn. We exclude simulated user language correctness because it is consistently above 0.95 across scenarios.

accessed via Azure AI Foundry. Table 8 summarizes the approximate total inference cost per scenario. For user model inference, we self-host Qwen3-235B-A22B-Instruct-2507 on 8×A100 GPUs (40GB each), resulting in a total of 4,032 GPU hours.

In the translation pipeline section, we used Gemini 3.1 Flash-Lite Preview for three domains and five target languages. Aggregated across five languages, the pipeline consumes 8.97M input tokens and 7.21M output tokens. The total one-time translation cost for three domains and five languages is approximately \$13.05.

E Agent language use and drift analysis

We characterize agent language behavior along three axes: how often the agent produces text in the expected target language (Figure 9), where in a conversation any off-target text appears (Figure 10), and how much of the off-target text is specifically English (Figure 11).

Correctness is high overall and degrades only in crosslingual interaction. Figure 9 shows that L2 Tool is essentially perfect across all tested languages (mean turn correctness ≈ 1.00 for Thai, Vietnamese, Indonesian, Chinese, and Filipino), indicating that translating tool schemas does not by itself induce agent language drift: with an English dialogue the agent reliably stays in English. L2 Domain also remains high (0.92–0.98), with a gentle ordering in which Thai and Vietnamese are strongest (0.98) and Filipino is weakest (0.92). The clearest degradation is confined to L2 Interaction, where correctness falls to 0.75–0.82 and is again



Figure 10: Proportion of non-L2 turns as a function of agent turn position, separated by setting and by speaker role (agent vs. user).

lowest for Filipino (0.75) and Chinese (0.78). The gap between settings is large relative to the gap between languages: moving from L2 Domain to L2 Interaction costs roughly 0.15–0.20 in correctness for every language, whereas the spread across languages within a setting is at most ~ 0.07 . Drift is therefore primarily a property of the interaction regime rather than of any single language.

Text of off-target language accumulates over the course of a conversation. Figure 10 decomposes this drift by turn position. In L2 Tool, the non-L2 proportion stays negligible throughout for both roles, consistent with scenario design where dialogue should be in English. In L2 Interaction and L2 Domain, the agent’s non-L2 share rises monotonically with turn index (Pearson $r \approx 0.65$ and $r \approx 0.73$ between turn position and off-target proportion, respectively), pointing to an accumulation or self-priming effect rather than a fixed per-turn error rate. The two settings differ in their dynamics: in L2 Interaction both the simulated user and the agent contribute off-target turns and the magnitude is largest (the agent share peaks near 0.12 in late turns), whereas in L2 Domain the simulated user remains essentially on-target (≈ 0.015 throughout) and the modest rise to ≈ 0.07 is agent-initiated and concentrated in the later turns. This is consistent with Figure 9: L2 Interaction is where both parties’ code-switching reinforces one another.

The off-target text is predominantly English, and only for Filipino is it non-trivial. Figure 11 shows that when the agent leaves the target language it overwhelmingly switches to English, but that this is rare for most languages. For Thai, Vietnamese, Indonesian, and Chinese the per-task median English share is 0 in both settings, with right-skewed distributions whose means sit at only a few percent. Filipino is the clear outlier: its English share is highest in both settings (mean 0.11, median 0.06 in L2 Interaction; mean 0.08, median 0.04 in L2 Domain), and the per-language mean

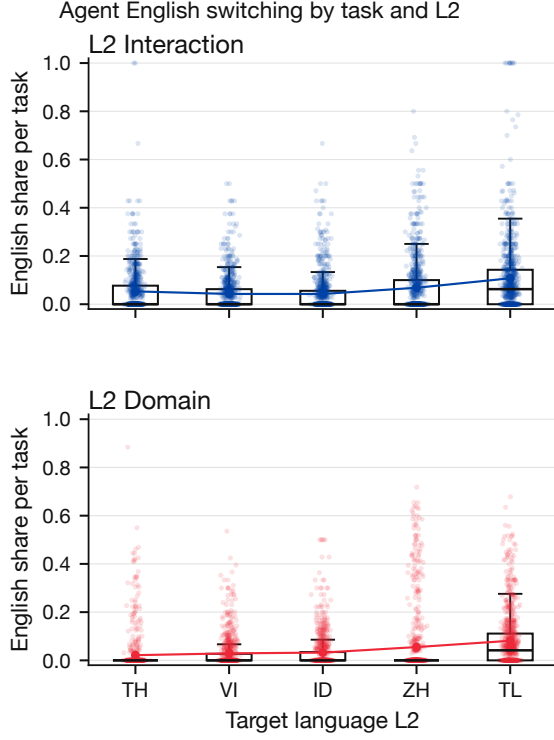


Figure 11: Per-task share of agent turns emitted in English, by target language, for the two scenarios in which the dialogue language is L2. Each dot is a task; the overlaid line tracks the per-language mean.

line rises toward Filipino in both panels. The right-skew indicates that English switching is driven by a minority of tasks rather than being uniform, so even the elevated Filipino mean reflects a tail of high-switching tasks rather than pervasive drift.

Drift is real but is not the dominant explanation for performance. Taken together, language drift is genuine and scenario-dependent: strongest in L2 Interaction, concentrated in Filipino, accumulating over turns, and realized mainly as English code-switching. It is nonetheless too localized and too weakly associated with task success to serve as the primary explanation for performance differences. Across all scenarios, agent language correctness is essentially uncorrelated with pass^3 (overall Pearson $r = -0.10$, $R^2 \approx 0.01$, $n = 166$); the only setting with a strong positive correlation is L2 Tool, where correctness is saturated near 1.00 and thus carries no usable variance, while in L2 Interaction and L2 Domain the scenario-level correlations are weak ($|r| \lesssim 0.25$). Language drift is therefore best read as a measurable secondary symptom of the harder crosslingual regimes (L2 Interaction) rather than a direct cause of the pass^3 gaps reported in the

main text.

F All results

Domain	Model	EN			
		p@1	p ²	p ³	ρ ³
Airline	GPT-5-mini	0.693	0.593	0.540	0.779
	Qwen3-235B	0.500	0.387	0.340	0.680
	Kimi-K2.5	0.707	0.600	0.540	0.764
Retail	GPT-5-mini	0.655	0.497	0.412	0.629
	Qwen3-235B	0.579	0.415	0.342	0.591
	Kimi-K2.5	0.561	0.442	0.395	0.704
Telecom	GPT-5-mini	0.713	0.588	0.509	0.714
	Qwen3-235B	0.357	0.205	0.132	0.370
	Kimi-K2.5	0.997	0.953	0.930	0.933

Table 9: English Baseline (S1) results. Where EN denotes English, p@1 denotes pass@1, p² denotes pass², and p³ denotes pass³. Bold values indicate the best score within each domain–metric group. Qwen3-235B refers to Qwen3-235B-A22B-Instruct-2507.

Domain	Model	VI				TH				ID				Zh				TL			
		p@1	p ²	p ³	ρ ³	p@1	p ²	p ³	ρ ³	p@1	p ²	p ³	ρ ³	p@1	p ²	p ³	ρ ³	p@1	p ²	p ³	ρ ³
Airline	GPT-5-mini	0.640	0.500	0.420	0.656	0.593	0.473	0.400	0.674	0.627	0.447	0.360	0.574	0.620	0.460	0.380	0.613	0.620	0.527	0.480	0.774
	Qwen3-235B	0.480	0.347	0.280	0.583	0.487	0.373	0.320	0.657	0.493	0.347	0.260	0.527	0.460	0.300	0.220	0.478	0.440	0.313	0.260	0.591
	Kimi-K2.5	0.733	0.660	0.600	0.818	0.687	0.580	0.520	0.757	0.740	0.653	0.600	0.811	0.760	0.640	0.580	0.763	0.747	0.653	0.600	0.804
Retail	GPT-5-mini	0.696	0.570	0.500	0.718	0.699	0.538	0.430	0.615	0.684	0.547	0.456	0.667	0.675	0.506	0.412	0.610	0.655	0.518	0.430	0.656
	Qwen3-235B	0.556	0.418	0.333	0.600	0.526	0.383	0.325	0.617	0.608	0.462	0.386	0.635	0.591	0.442	0.351	0.594	0.573	0.444	0.386	0.674
	Kimi-K2.5	0.687	0.550	0.465	0.677	0.573	0.430	0.360	0.627	0.640	0.483	0.386	0.603	0.722	0.597	0.526	0.729	0.675	0.541	0.474	0.701
Telecom	GPT-5-mini	0.561	0.430	0.368	0.656	0.605	0.436	0.342	0.565	0.582	0.453	0.404	0.693	0.728	0.608	0.518	0.711	0.740	0.617	0.535	0.723
	Qwen3-235B	0.260	0.129	0.061	0.236	0.222	0.108	0.061	0.276	0.333	0.170	0.105	0.316	0.228	0.132	0.097	0.423	0.155	0.088	0.061	0.396
	Kimi-K2.5	0.912	0.836	0.772	0.846	0.854	0.763	0.702	0.822	0.915	0.839	0.772	0.843	0.930	0.868	0.816	0.877	0.883	0.784	0.702	0.795

Table 10: L2 Interaction (S2) results across Vietnamese (VI), Thai (TH), Indonesian (ID), Chinese (ZH), and Filipino (TL). Here, p@1 denotes pass@1, p² denotes pass², and p³ denotes pass³. Bold values indicate the best score within each domain–metric group. Qwen3-235B refers to Qwen3-235B-A22B-Instruct-2507.

Domain	Model	VI				TH				ID				Zh				TL			
		p@1	p ²	p ³	ρ ³	p@1	p ²	p ³	ρ ³	p@1	p ²	p ³	ρ ³	p@1	p ²	p ³	ρ ³	p@1	p ²	p ³	ρ ³
Airline	GPT-5-mini	0.653	0.527	0.460	0.704	0.620	0.473	0.400	0.645	0.687	0.567	0.480	0.699	0.687	0.593	0.520	0.757	0.667	0.580	0.540	0.810
	Qwen3-235B	0.447	0.313	0.260	0.582	0.380	0.253	0.220	0.579	0.400	0.233	0.180	0.450	0.480	0.313	0.200	0.417	0.467	0.320	0.240	0.514
Retail	GPT-5-mini	0.681	0.532	0.430	0.631	0.646	0.477	0.377	0.584	0.675	0.526	0.430	0.637	0.719	0.579	0.482	0.670	0.661	0.515	0.439	0.664
	Qwen3-235B	0.547	0.412	0.351	0.642	0.561	0.421	0.351	0.626	0.550	0.418	0.351	0.638	0.594	0.433	0.360	0.606	0.570	0.421	0.342	0.600
Telecom	GPT-5-mini	0.664	0.550	0.482	0.726	0.713	0.588	0.518	0.727	0.713	0.596	0.518	0.727	0.705	0.576	0.500	0.709	0.719	0.594	0.509	0.708
	Qwen3-235B	0.254	0.135	0.088	0.346	0.339	0.208	0.140	0.413	0.269	0.155	0.105	0.390	0.222	0.135	0.096	0.432	0.301	0.167	0.105	0.349

Table 11: Tool Adaptation (S3) results in monolingual settings across Vietnamese (VI), Thai (TH), Indonesian (ID), Chinese (ZH), and Filipino (TL). Here, p@1 denotes pass@1, p² denotes pass², and p³ denotes pass³. Bold values indicate the best score within each domain–metric group. Qwen3-235B refers to Qwen3-235B-A22B-Instruct-2507.

Domain	Model	Bi				Tri				Quad				Multi			
		p@1	p ²	p ³	ρ ³	p@1	p ²	p ³	ρ ³	p@1	p ²	p ³	ρ ³	p@1	p ²	p ³	ρ ³
Airline	GPT-5-mini	0.680	0.560	0.480	0.706	0.660	0.560	0.480	0.727	0.640	0.520	0.480	0.750	0.640	0.513	0.440	0.688
	Qwen3-235B	0.440	0.280	0.200	0.455	0.413	0.267	0.200	0.484	0.413	0.287	0.240	0.581	0.413	0.267	0.220	0.533
Retail	GPT-5-mini	0.667	0.541	0.465	0.697	0.667	0.538	0.456	0.684	0.652	0.485	0.386	0.592	0.696	0.547	0.447	0.642
	Qwen3-235B	0.596	0.456	0.386	0.648	0.535	0.398	0.325	0.607	0.585	0.447	0.377	0.644	0.570	0.401	0.298	0.523
Telecom	GPT-5-mini	0.664	0.541	0.482	0.726	0.696	0.558	0.491	0.705	0.670	0.550	0.491	0.733	0.652	0.520	0.465	0.713
	Qwen3-235B	0.298	0.155	0.088	0.295	0.392	0.213	0.140	0.357	0.365	0.181	0.105	0.288	0.342	0.181	0.114	0.333

Table 12: Tool Adaptation (S3) results for multilingual settings, including bilingual (Bi), trilingual (Tri), quadlingual (Quad), and multilingual (Multi) configurations. Here, p@1 denotes pass@1, p² denotes pass², and p³ denotes pass³. Bold values indicate the best score within each domain–metric group. Qwen3-235B refers to Qwen3-235B-A22B-Instruct-2507.

Domain	Model	VI				TH				ID				Zh				TL			
		p@1	p ²	p ³	ρ^3	p@1	p ²	p ³	ρ^3	p@1	p ²	p ³	ρ^3	p@1	p ²	p ³	ρ^3	p@1	p ²	p ³	ρ^3
Airline	GPT-5-mini	0.540	0.400	0.320	0.593	0.593	0.453	0.360	0.607	0.600	0.540	0.500	0.833	0.607	0.493	0.440	0.725	0.560	0.433	0.380	0.679
	Qwen3-235B	0.487	0.460	0.440	0.903	0.460	0.413	0.380	0.826	0.480	0.440	0.420	0.875	0.500	0.473	0.460	0.920	0.473	0.420	0.400	0.846
	Kimi-K2.5	0.560	0.427	0.340	0.607	0.547	0.387	0.300	0.548	0.600	0.487	0.420	0.700	0.660	0.507	0.440	0.667	0.600	0.487	0.400	0.667
Retail	GPT-5-mini	0.585	0.427	0.333	0.569	0.325	0.175	0.123	0.378	0.541	0.392	0.325	0.601	0.596	0.450	0.368	0.617	0.488	0.333	0.263	0.539
	Qwen3-235B	0.289	0.205	0.149	0.516	0.178	0.143	0.132	0.742	0.243	0.170	0.140	0.576	0.266	0.181	0.149	0.560	0.260	0.187	0.158	0.608
	Kimi-K2.5	0.567	0.392	0.268	0.473	0.327	0.190	0.114	0.349	0.433	0.287	0.202	0.467	0.607	0.450	0.366	0.603	0.444	0.292	0.219	0.493
Telecom	GPT-5-mini	0.564	0.427	0.360	0.638	0.599	0.447	0.360	0.601	0.497	0.325	0.246	0.495	0.760	0.652	0.588	0.774	0.681	0.547	0.474	0.696
	Qwen3-235B	0.383	0.213	0.149	0.389	0.284	0.143	0.079	0.278	0.427	0.295	0.211	0.494	0.330	0.205	0.149	0.452	0.330	0.205	0.149	0.452
	Kimi-K2.5	0.699	0.529	0.404	0.578	0.693	0.523	0.43	0.620	0.798	0.681	0.588	0.737	0.863	0.763	0.684	0.793	0.743	0.582	0.474	0.638

Table 13: L2 Domain (S4) results across Vietnamese (VI), Thai (TH), Indonesian (ID), Chinese (ZH), and Filipino (TL). Here, p@1 denotes pass@1, p² denotes pass², and p³ denotes pass³. Bold values indicate the best score within each domain–metric group. Qwen3-235B refers to Qwen3-235B-A22B-Instruct-2507.

G Detailed Error Analysis Results

Setting	Variant	Agent Errors			User Errors		
		Critical	Minor	Correct	Critical	Minor	Correct
English Baseline	English	81	22	47	38	48	64
L2 Interaction	Vietnamese	93	25	32	43	48	59
	Thai	105	14	31	42	45	63
	Indonesian	88	18	44	36	49	65
	Chinese	90	18	42	44	51	55
	Filipino	94	18	38	53	36	61
Tool Adaptation	VI Tools	81	23	46	43	51	56
	TH Tools	88	28	34	39	45	66
	ID Tools	94	20	36	41	41	68
	ZH Tools	84	23	43	35	44	71
	FIL Tools	92	17	41	45	45	60
	Mix-2 (EN+TH)	90	24	36	52	38	60
	Mix-3 (+VI)	86	17	47	33	54	63
	Mix-4 (+ID)	89	26	35	35	45	70
	Mix-5 (+ZH)	102	15	33	40	41	69
L2 Domain	Vietnamese	83	27	40	43	41	66
	Thai	97	17	36	55	30	65
	Indonesian	92	24	34	42	39	69
	Chinese	103	15	32	40	48	62
	Filipino	101	17	32	50	47	53

Table 14: Error analysis on the **Airline** domain (gpt-5-mini agent, qwen3-235b user simulator, DeepSeek-V4-Flash judge, $n=150$ per condition). Counts of simulations by maximum severity level.

Setting	Variant	Agent Errors			User Errors		
		Critical	Minor	Correct	Critical	Minor	Correct
English Baseline	English	98	70	174	79	135	128
L2 Interaction	Vietnamese	109	78	155	96	103	143
	Thai	125	73	144	94	124	124
	Indonesian	117	65	160	86	121	135
	Chinese	109	73	160	81	126	135
	Filipino	103	88	151	105	129	108
Tool Adaptation	VI Tools	97	75	170	75	130	137
	TH Tools	102	79	161	70	123	149
	ID Tools	94	86	162	92	130	120
	ZH Tools	95	64	183	87	121	134
	FIL Tools	88	79	175	79	129	134
	Mix-2 (EN+TH)	81	81	180	58	132	152
	Mix-3 (+VI)	100	69	173	88	123	131
	Mix-4 (+ID)	100	72	170	88	112	142
	Mix-5 (+ZH)	87	93	162	82	129	131
L2 Domain	Vietnamese	112	93	137	90	122	130
	Thai	180	60	102	127	91	124
	Indonesian	124	76	142	93	122	127
	Chinese	131	84	127	101	113	128
	Filipino	131	74	137	123	118	101

Table 15: Error analysis on the **Retail** domain (gpt-5-mini agent, qwen3-235b user simulator, DeepSeek-V4-Flash judge, $n=342$ per condition). Counts of simulations by maximum severity level.

Setting	Variant	Agent Errors			User Errors		
		Critical	Minor	Correct	Critical	Minor	Correct
English Baseline	English	110	113	119	50	130	162
L2 Interaction	Vietnamese	101	116	125	52	146	144
	Thai	125	90	127	59	133	150
	Indonesian	122	100	120	60	136	146
	Chinese	95	76	171	60	129	153
	Filipino	91	88	163	62	141	139
Tool Adaptation	VI Tools	103	117	122	45	156	141
	TH Tools	87	106	149	46	147	149
	ID Tools	89	108	145	48	140	154
	ZH Tools	109	91	142	55	135	152
	FIL Tools	75	112	155	45	115	182
	Mix-2 (EN+TH)	91	106	145	54	138	150
	Mix-3 (+VI)	95	117	130	55	144	143
	Mix-4 (+ID)	91	102	149	56	128	158
	Mix-5 (+ZH)	94	101	147	43	148	151
L2 Domain	Vietnamese	125	71	146	54	112	176
	Thai	133	85	124	64	105	173
	Indonesian	139	81	122	59	120	163
	Chinese	103	86	153	45	128	169
	Filipino	131	89	122	64	130	148

Table 16: Error analysis on the **Telecom** domain (gpt-5-mini agent, qwen3-235b user simulator, DeepSeek-V4-Flash judge, $n=342$ per condition). Counts of simulations by maximum severity level.

Error Tag	Eng. Baseline	L2 Interaction					Tool Adaptation					L2 Domain								
	EN	VI	TH	ID	ZH	FIL	VI	TH	ID	ZH	FIL	Mix-2	Mix-3	Mix-4	Mix-5	VI	TH	ID	ZH	FIL
Guideline Violation	C:100 M:20	C:116 M:38	C:113 M:23	C:89 M:31	C:85 M:27	C:110 M:28	C:82 M:30	C:87 M:32	C:92 M:26	C:89 M:21	C:110 M:22	C:97 M:16	C:96 M:24	C:103 M:26	C:90 M:26	C:130 M:28	C:115 M:18	C:108 M:21	C:108 M:21	C:114 M:24
Hallucination	C:6 M:7	C:3 M:6	C:4 M:8	C:6 M:5	C:9 M:1	C:3 M:5	C:0 M:4	C:5 M:2	C:2 M:4	C:6 M:5	C:3 M:2	C:3 M:5	C:7 M:2	C:3 M:5	C:7 M:2	C:8 M:2	C:14 M:10	C:3 M:5	C:7 M:8	C:6 M:5
Inconsistent Behavior	C:4 M:8	C:7 M:11	C:11 M:9	C:6 M:15	C:5 M:7	C:12 M:5	C:11 M:3	C:5 M:6	C:5 M:10	C:3 M:5	C:13 M:4	C:8 M:9	C:6 M:5	C:6 M:13	C:9 M:10	C:6 M:8	C:7 M:10	C:6 M:5	C:7 M:10	C:6 M:5
Incorrect Interpretation	C:33 M:25	C:67 M:29	C:56 M:30	C:47 M:33	C:46 M:38	C:46 M:36	C:25 M:29	C:46 M:35	C:33 M:40	C:42 M:31	C:39 M:29	C:34 M:29	C:33 M:24	C:36 M:34	C:46 M:21	C:54 M:37	C:71 M:35	C:43 M:25	C:47 M:33	C:58 M:30
Interruption Error	C:0 M:1	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:1	C:0 M:1	C:0 M:0
Irrelevant Tool Call	C:0 M:0	C:0 M:0	C:0 M:4	C:0 M:4	C:0 M:2	C:0 M:0	C:0 M:0	C:0 M:1	C:0 M:0	C:0 M:2	C:0 M:1	C:1 M:1	C:1 M:3	C:0 M:2	C:0 M:1	C:1 M:4	C:1 M:0	C:0 M:1	C:0 M:1	C:0 M:1
Missed Required Action	C:71 M:16	C:82 M:28	C:98 M:18	C:74 M:15	C:85 M:20	C:90 M:25	C:79 M:22	C:78 M:22	C:78 M:17	C:75 M:27	C:85 M:22	C:78 M:20	C:78 M:14	C:87 M:33	C:84 M:21	C:77 M:15	C:86 M:16	C:96 M:25	C:94 M:29	C:110 M:23
Other	C:0 M:1	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0
Premature Termination	C:28 M:1	C:30 M:1	C:30 M:2	C:21 M:1	C:18 M:0	C:25 M:1	C:22 M:1	C:21 M:4	C:19 M:13	C:23 M:3	C:34 M:1	C:29 M:1	C:35 M:2	C:26 M:1	C:45 M:2	C:21 M:9	C:25 M:2	C:28 M:0	C:27 M:2	C:30 M:1
Revealed Info Early	C:0 M:1	C:0 M:2	C:0 M:1	C:1 M:0	C:1 M:0	C:2 M:2	C:3 M:1	C:1 M:1	C:0 M:0	C:0 M:0	C:1 M:1	C:2 M:1	C:0 M:0	C:0 M:0	C:1 M:2	C:0 M:2	C:0 M:2	C:1 M:2	C:0 M:0	C:0 M:0
Tool Call Argument Error	C:5 M:4	C:3 M:1	C:7 M:6	C:4 M:6	C:3 M:7	C:1 M:4	C:2 M:6	C:2 M:6	C:1 M:4	C:1 M:0	C:5 M:0	C:3 M:3	C:2 M:5	C:1 M:4	C:1 M:2	C:5 M:3	C:3 M:3	C:7 M:8	C:6 M:4	C:6 M:2
Tool Call Schema Error	C:0 M:0	C:1 M:0	C:2 M:2	C:0 M:0	C:0 M:1	C:0 M:4	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:1	C:1 M:0	C:0 M:1	C:0 M:2	C:1 M:1	C:0 M:0	C:0 M:1	C:0 M:1	C:0 M:1	C:0 M:0
Wrong Sequence	C:9 M:3	C:9 M:7	C:8 M:3	C:3 M:4	C:11 M:4	C:7 M:3	C:12 M:6	C:3 M:3	C:5 M:3	C:12 M:6	C:5 M:2	C:5 M:1	C:11 M:1	C:5 M:1	C:7 M:1	C:8 M:3	C:7 M:3	C:13 M:5	C:12 M:2	C:11 M:5

Table 17: Agent error tag counts — all settings, **Airline** domain (gpt-5-mini agent, qwen3-235b user simulator, DeepSeek-V4-Flash judge, $n=150$ per condition). **C:** = critical-severity count; **M:** = minor-severity count. Each value is the number of simulations in which the tag appeared at that severity level at least once.

Error Tag	Eng. Baseline	L2 Interaction					Tool Adaptation					L2 Domain								
	EN	VI	TH	ID	ZH	FIL	VI	TH	ID	ZH	FIL	Mix-2	Mix-3	Mix-4	Mix-5	VI	TH	ID	ZH	FIL
Guideline Violation	C:13 M:26	C:18 M:22	C:19 M:25	C:15 M:30	C:28 M:23	C:26 M:31	C:19 M:36	C:9 M:17	C:16 M:13	C:14 M:25	C:22 M:18	C:23 M:19	C:16 M:22	C:19 M:20	C:18 M:14	C:16 M:20	C:29 M:14	C:20 M:13	C:18 M:19	C:21 M:24
Hallucination	C:26 M:27	C:22 M:24	C:31 M:24	C:19 M:37	C:23 M:31	C:39 M:23	C:39 M:28	C:39 M:20	C:25 M:21	C:32 M:28	C:35 M:29	C:46 M:19	C:30 M:26	C:25 M:21	C:33 M:20	C:35 M:22	C:48 M:13	C:24 M:13	C:33 M:18	C:29 M:21
Inconsistent Behavior	C:15 M:34	C:17 M:24	C:15 M:28	C:17 M:42	C:25 M:35	C:25 M:31	C:15 M:34	C:19 M:27	C:13 M:40	C:14 M:37	C:24 M:33	C:21 M:37	C:19 M:36	C:15 M:37	C:25 M:33	C:16 M:23	C:26 M:30	C:15 M:27	C:18 M:37	C:19 M:39
Incorrect Interpretation	C:4 M:7	C:3 M:8	C:2 M:9	C:2 M:4	C:3 M:9	C:11 M:7	C:2 M:11	C:3 M:2	C:3 M:5	C:1 M:6	C:1 M:4	C:6 M:8	C:4 M:9	C:3 M:4	C:1 M:7	C:3 M:8	C:3 M:2	C:3 M:6	C:10 M:15	C:5 M:8
Interruption Error	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0
Irrelevant Tool Call	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0
Missed Required Action	C:0 M:1	C:3 M:1	C:2 M:0	C:0 M:1	C:1 M:2	C:2 M:1	C:1 M:1	C:1 M:0	C:3 M:1	C:1 M:0	C:1 M:2	C:1 M:1	C:0 M:0	C:1 M:3	C:0 M:1	C:2 M:3	C:0 M:1	C:0 M:0	C:0 M:0	C:3 M:3
Other	C:0 M:1	C:0 M:1	C:0 M:0	C:0 M:0	C:0 M:2	C:1 M:0	C:0 M:0	C:0 M:2	C:0 M:0	C:1 M:0	C:0 M:0	C:0 M:0	C:0 M:3	C:1 M:0	C:0 M:0	C:0 M:2	C:0 M:1	C:0 M:1	C:0 M:0	C:2 M:1
Premature Termination	C:9 M:10	C:6 M:12	C:12 M:15	C:11 M:9	C:11 M:18	C:8 M:10	C:4 M:7	C:10 M:17	C:10 M:16	C:7 M:10	C:9 M:15	C:9 M:7	C:7 M:9	C:6 M:11	C:3 M:11	C:8 M:14	C:10 M:8	C:9 M:11	C:9 M:12	C:11 M:16
Revealed Info Early	C:0 M:2	C:1 M:0	C:0 M:2	C:0 M:0	C:0 M:1	C:0 M:2	C:0 M:2	C:0 M:0	C:0 M:0	C:1 M:1	C:0 M:1	C:0 M:0	C:0 M:0	C:0 M:1	C:0 M:3	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:3	C:0 M:1
Tool Call Argument Error	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0
Tool Call Schema Error	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0
Wrong Sequence	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:1	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:1	C:0 M:1	C:0 M:0	C:1 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:1	C:0 M:0

Table 18: User error tag counts — all settings, **Airline** domain (gpt-5-mini agent, qwen3-235b user simulator, DeepSeek-V4-Flash judge, $n=150$ per condition). **C:** = critical-severity count; **M:** = minor-severity count. Each value is the number of simulations in which the tag appeared at that severity level at least once.

Error Tag	Eng. Baseline		L2 Interaction					Tool Adaptation					L2 Domain							
	EN		TH	ID	ZH	FIL	VI	TH	ID	ZH	FIL	Mix-2	Mix-3	Mix-4	Mix-5	VI	TH	ID	ZH	FIL
	VI		TH	ID	ZH	FIL	VI	TH	ID	ZH	FIL	Mix-2	Mix-3	Mix-4	Mix-5	VI	TH	ID	ZH	FIL
Guideline Violation	C:70 M:74	C:99 M:80	C:105 M:81	C:88 M:83	C:71 M:59	C:86 M:94	C:68 M:52	C:70 M:66	C:69 M:69	C:62 M:55	C:78 M:56	C:69 M:55	C:67 M:48	C:63 M:53	C:78 M:72	C:73 M:76	C:128 M:54	C:92 M:53	C:112 M:93	C:92 M:59
Hallucination	C:7 M:11	C:9 M:12	C:15 M:13	C:17 M:17	C:13 M:11	C:10 M:10	C:9 M:11	C:6 M:15	C:5 M:11	C:16 M:7	C:15 M:10	C:11 M:10	C:11 M:12	C:17 M:18	C:12 M:18	C:18 M:17	C:30 M:10	C:18 M:17	C:9 M:18	C:14 M:13
Inconsistent Behavior	C:10 M:15	C:17 M:24	C:13 M:24	C:8 M:15	C:8 M:28	C:8 M:16	C:10 M:15	C:6 M:14	C:7 M:17	C:8 M:10	C:5 M:21	C:10 M:13	C:7 M:16	C:8 M:26	C:13 M:27	C:29 M:25	C:24 M:27	C:22 M:22	C:22 M:22	C:22 M:22
Incorrect Interpretation	C:34 M:39	C:32 M:54	C:42 M:59	C:38 M:54	C:26 M:44	C:37 M:39	C:31 M:53	C:40 M:47	C:28 M:39	C:30 M:40	C:36 M:34	C:22 M:33	C:29 M:32	C:38 M:43	C:27 M:65	C:48 M:88	C:142 M:62	C:50 M:56	C:63 M:63	C:50 M:55
Interruption Error	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0
Irrelevant Tool Call	C:2 M:3	C:1 M:3	C:1 M:4	C:5 M:5	C:0 M:5	C:0 M:2	C:1 M:3	C:1 M:5	C:0 M:0	C:0 M:2	C:2 M:2	C:0 M:1	C:1 M:3	C:0 M:3	C:1 M:5	C:1 M:1	C:1 M:4	C:1 M:5	C:1 M:1	C:1 M:1
Missed Required Action	C:75 M:53	C:78 M:62	C:95 M:51	C:88 M:55	C:94 M:63	C:62 M:59	C:79 M:66	C:74 M:64	C:69 M:63	C:69 M:69	C:58 M:56	C:63 M:71	C:73 M:49	C:59 M:67	C:73 M:66	C:160 M:62	C:75 M:65	C:98 M:83	C:101 M:81	C:101 M:81
Other	C:0 M:2	C:0 M:3	C:1 M:7	C:2 M:1	C:0 M:3	C:0 M:3	C:0 M:3	C:0 M:3	C:2 M:1	C:1 M:2	C:1 M:2	C:0 M:4	C:0 M:1	C:0 M:6	C:0 M:3	C:1 M:5	C:1 M:5	C:1 M:2	C:1 M:5	C:2 M:1
Premature Termination	C:0 M:5	C:0 M:5	C:0 M:5	C:0 M:5	C:0 M:5	C:0 M:5	C:0 M:5	C:0 M:5	C:0 M:5	C:0 M:5	C:0 M:5	C:0 M:5	C:0 M:5	C:0 M:5	C:0 M:5	C:0 M:5	C:0 M:5	C:0 M:5	C:0 M:5	C:0 M:5
Revealed Info Early	C:0 M:0	C:1 M:5	C:4 M:4	C:4 M:2	C:1 M:2	C:2 M:3	C:2 M:3	C:1 M:2	C:1 M:5	C:0 M:6	C:0 M:0	C:0 M:4	C:3 M:1	C:0 M:5	C:1 M:0	C:1 M:3	C:2 M:3	C:2 M:3	C:0 M:2	C:2 M:2
Tool Call Argument Error	C:8 M:5	C:7 M:7	C:7 M:18	C:8 M:6	C:9 M:10	C:5 M:3	C:9 M:7	C:8 M:9	C:6 M:3	C:8 M:6	C:5 M:14	C:7 M:4	C:8 M:9	C:5 M:8	C:35 M:49	C:48 M:39	C:39 M:28	C:48 M:25	C:37 M:34	C:37 M:34
Tool Call Schema Error	C:1 M:2	C:3 M:5	C:4 M:6	C:1 M:5	C:0 M:4	C:4 M:9	C:0 M:6	C:2 M:6	C:1 M:8	C:4 M:2	C:4 M:5	C:1 M:2	C:0 M:9	C:4 M:5	C:2 M:7	C:6 M:1	C:3 M:9	C:5 M:9	C:6 M:5	C:2 M:4
Wrong Sequence	C:20 M:7	C:24 M:24	C:20 M:17	C:19 M:20	C:19 M:16	C:23 M:15	C:16 M:14	C:12 M:21	C:18 M:18	C:8 M:13	C:11 M:9	C:14 M:14	C:7 M:10	C:11 M:13	C:14 M:11	C:24 M:18	C:19 M:21	C:12 M:15	C:13 M:14	C:13 M:14

Table 19: Agent error tag counts — all settings, **Retail** domain (gpt-5-mini agent, qwen3-235b user simulator, DeepSeek-V4-Flash judge, $n=342$ per condition). **C:** = critical-severity count; **M:** = minor-severity count. Each value is the number of simulations in which the tag appeared at that severity level at least once.

Error Tag	Eng. Baseline	L2 Interaction										Tool Adaptation										L2 Domain																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																		
	EN	VI					TH					ID					ZH					FIL					Mix-2					Mix-3					Mix-4					Mix-5					VI					TH					ID					ZH					FIL																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																					

Error Tag	Eng. Baseline	L2 Interaction					Tool Adaptation										L2 Domain				
	EN	VI	TH	ID	ZH	FIL	VI	TH	ID	ZH	FIL	Mix-2	Mix-3	Mix-4	Mix-5	VI	TH	ID	ZH	FIL	
Guideline Violation	C:25 M:42	C:10 M:93	C:30 M:77	C:30 M:53	C:30 M:82	C:54 M:105	C:33 M:52	C:36 M:61	C:16 M:74	C:19 M:64	C:35 M:58	C:15 M:49	C:13 M:52	C:15 M:77	C:12 M:56	C:13 M:42	C:17 M:46	C:11 M:59	C:23 M:85	C:22 M:85	
Hallucination	C:52 M:85	C:34 M:96	C:47 M:67	C:68 M:74	C:52 M:85	C:29 M:68	C:45 M:91	C:48 M:88	C:51 M:78	C:49 M:55	C:32 M:60	C:48 M:93	C:40 M:95	C:38 M:69	C:74 M:83	C:39 M:48	C:49 M:56	C:57 M:54	C:37 M:61	C:52 M:47	
Inconsistent Behavior	C:24 M:194	C:19 M:207	C:26 M:186	C:33 M:139	C:40 M:197	C:52 M:194	C:17 M:186	C:47 M:175	C:15 M:180	C:22 M:160	C:23 M:135	C:28 M:158	C:32 M:209	C:23 M:169	C:14 M:184	C:18 M:146	C:16 M:151	C:20 M:140	C:10 M:115	C:20 M:142	
Incorrect Interpretation	C:1 M:50	C:6 M:30	C:2 M:36	C:8 M:49	C:7 M:38	C:3 M:31	C:4 M:30	C:6 M:36	C:5 M:49	C:6 M:52	C:4 M:46	C:4 M:38	C:7 M:35	C:4 M:39	C:4 M:29	C:3 M:30	C:7 M:34	C:8 M:25	C:6 M:28	C:7 M:42	
Interruption Error	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	C:0 M:0	
Irrelevant Tool Call	C:0 M:0	C:0 M:1	C:0 M:1	C:0 M:4	C:0 M:0	C:0 M:0	C:0 M:1	C:0 M:1	C:0 M:8	C:0 M:3	C:0 M:10	C:0 M:0	C:0 M:1	C:0 M:3	C:0 M:0	C:0 M:1	C:0 M:0	C:0 M:1	C:0 M:3	C:0 M:3	
Missed Required Action	C:3 M:24	C:3 M:9	C:6 M:16	C:1 M:22	C:3 M:12	C:2 M:29	C:2 M:11	C:2 M:30	C:2 M:21	C:5 M:16	C:9 M:11	C:3 M:22	C:2 M:8	C:4 M:17	C:1 M:21	C:6 M:8	C:2 M:12	C:1 M:9	C:3 M:14	C:2 M:9	
Other	C:1 M:10	C:1 M:9	C:0 M:13	C:1 M:9	C:0 M:28	C:0 M:15	C:0 M:15	C:1 M:3	C:0 M:12	C:0 M:32	C:0 M:11	C:1 M:9	C:0 M:16	C:1 M:10	C:1 M:10	C:1 M:11	C:0 M:33	C:0 M:20	C:1 M:8	C:0 M:27	
Premature Termination	C:13 M:16	C:9 M:24	C:12 M:20	C:10 M:18	C:14 M:21	C:17 M:16	C:5 M:23	C:3 M:14	C:12 M:20	C:8 M:15	C:11 M:14	C:11 M:15	C:12 M:17	C:15 M:14	C:5 M:16	C:11 M:23	C:17 M:12	C:14 M:28	C:9 M:18	C:11 M:31	
Revealed Info Early	C:0 M:0	C:0 M:0	C:0 M:2	C:0 M:1	C:1 M:3	C:0 M:3	C:0 M:1	C:0 M:0	C:0 M:4	C:0 M:15	C:1 M:1	C:1 M:1	C:0 M:0	C:0 M:0	C:1 M:0	C:1 M:1	C:1 M:2	C:0 M:2	C:0 M:1	C:0 M:1	
Tool Call Argument Error	C:0 M:3	C:2 M:7	C:0 M:2	C:2 M:1	C:1 M:0	C:1 M:2	C:0 M:10	C:0 M:5	C:1 M:9	C:0 M:1	C:0 M:2	C:1 M:4	C:4 M:8	C:0 M:1	C:0 M:1	C:0 M:5	C:0 M:2	C:0 M:1	C:0 M:3	C:0 M:2	
Tool Call Schema Error	C:0 M:1	C:5 M:6	C:0 M:0	C:0 M:2	C:0 M:8	C:0 M:5	C:0 M:3	C:0 M:6	C:0 M:5	C:0 M:2	C:0 M:4	C:0 M:3	C:0 M:1	C:0 M:5	C:0 M:1	C:0 M:0	C:0 M:1	C:0 M:22	C:1 M:1	C:0 M:2	
Wrong Sequence	C:0 M:0	C:0 M:0	C:0 M:11	C:4 M:6	C:2 M:5	C:0 M:4	C:0 M:7	C:1 M:2	C:0 M:6	C:2 M:3	C:0 M:3	C:0 M:10	C:0 M:1	C:0 M:5	C:0 M:2	C:0 M:2	C:1 M:1	C:0 M:1	C:0 M:1	C:0 M:1	

Table 22: User error tag counts — all settings, **Telecom** domain (gpt-5-mini agent, qwen3-235b user simulator, DeepSeek-V4-Flash judge, $n=342$ per condition). **C:** = critical-severity count; **M:** = minor-severity count. Each value is the number of simulations in which the tag appeared at that severity level at least once.